

Carbon-fiber tape

Carbon-fiber tape is a flat material made of carbon fiber. It weighs one-seventh as much as steel for a given strength. The carbon fiber core lasts longer than conventional steel cable. The material is resistant to wear and abrasion and, unlike steel, does not densify and stretch.^[1]

Applications

In June 2013, KONE elevator company announced Ultrarope for use as a replacement for steel cables in elevators. It seals the carbon fibers in high-friction polymer. Unlike steel cable, Ultrarope was designed for buildings that require up to 1,000 meters of lift. Steel elevators top out at 500 meters.

The company estimated that in a 500-meter-high building, an elevator would use 15 per cent less electrical power than a steel-cabled version. As of June 2013, the product had passed all European Union and US certification tests.^[1]

See also

- Carbon fibers
- Carbon nanotube

References

1. "UltraRope announced to one-stop zoom up tall buildings" (<http://phys.org/news/2013-06-ultraropec-one-stop-tall.html>). Phys.org. Retrieved 2013-06-13.

External links

- Carbon Fiber Products (<http://www.bestwillcomposite.com/>)
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